Universal Surgical Alignment

Leveraging Canonical References for Zero-Shot Workflow Recognition

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Primary Research Question

The Scope (Refined & Streamlined)

We eliminate manual, labor-intensive video captioning and text cataloging pipelines. Instead, we leverage pre-existing, standard canonical checklists from surgical guidelines and clinical drapes.

Core Thesis Question

Can we leverage **hierarchical video-language alignment** to automatically synchronize raw surgical video streams with predefined canonical step checklists to enable **zero-shot procedural recognition** and maintain tracking continuity when steps are skipped or performed out of sequence?

Why Universal Alignment Matters

Surgical workflow recognition is the foundation for next-generation Computer-Assisted Surgery (CAS) systems:

- **Clinical Checkpoints:** Establish progress tracking inside the operating theatre to coordinate surgical teams.
- **Milestone Verification:** Confirm in real-time if crucial clinical milestones are successfully reached.
- **Safety Guardrails:** Automatically flag bypassed steps or missing safety checkpoints to prevent intraoperative adverse events.

Phases of Surgery: Procedural Anatomy

Surgical procedures are modular sequences of drapes and tasks that can be generalized into four macro-phases:

1. Access

Gaining entry to the target site. Includes drapes, markings, incisions, and viscodilatation.

2. Remove/Repair/Reconstruct

The core therapeutic task (e.g., lens emulsification, tissue resection).

3. Secure

Ensuring structural integrity (e.g., implant positioning, clip placement, suturing).

4. Close & Verify

Closing the surgical field and checking for leaks (e.g., wound sealing control).

Multi-Level Semantic Scene Understanding (MSSU)

Surgical scenes are complex drapes of objects, tools, and actions. Following **HCT+ (MICCAI 2024)**, we decompose procedures hierarchically:

Phase $\rightarrow$ **Step** $\rightarrow$ **Action** $\rightarrow$ **Instrument** $\rightarrow$ **Anatomy**

Hierarchical Complementarity Example

- **Phase:** Opening
- **Step:** Incision
- **Action:** Cut
- **Instrument:** Primary Knife
- **Anatomy:** Corneal Limbus

NLP Analogy: Similar to mapping Intents, Actions, and Entities in conversational dialogue flows.

Action-Driven Digital Twins (ActDTs)

To avoid visual token compression and enable semantic reasoning, we serialize video frames as structured, spatio-temporal **JSON triplets** representing Subject-Action-Object relations:

- **Object Segmentation:** Generated by a custom Mask2Former backbone.
- **3D Bounding Boxes:** Combines 2D coordinates with DepthAnythingV3 centroid depths (x, y, z).
- **Action Classification:** Extracted frame-by-frame via Qwen3-VL-8B.


```
{
  "interval": [120.0, 145.0],
  "actions": [
    {
      "subject": "Primary Knife",
      "action": "Cut",
      "object": "Cornea",
      "loc_3D": [342, 198, 480, 256, 1.45]
    }
  ]
}
```

Reasoning Retrieval via Digital Twins: OR^3

Surgical video retrieval requires reasoning over implicit queries (e.g., *"the step right before clipping"*) across visually near-identical frames.

- **The Bottleneck:** Global embedding matches (CLIP4Clip, X-CLIP) fail because sequential clips share identical drapes, drapes, and lighting.
- **The OR^3 Solution (Hopkins, 2025):**
  - Uses an LLM to "imagine" a hypothetical ActDT from an implicit query.
  - Performs intra-modal matching (text-to-text JSON structure) to bridge the cross-modal semantic gap.
  - Employs evidence-grounded refinement to iteratively align clips with clinical evidence.

Multi-Granular Zero-Shot Alignment: HecVL

To bypass manual frame-labeling drapes, we optimize a shared visual-linguistic encoder across distinct, disentangled embedding spaces:

- **Snarration (Clip-level):** Atomic visual features aligned with short-term narration transcripts.
- **Sconcept (Phase-level):** Aggregated clip features matched with conceptual procedural summaries.
- **Sabstract (Video-level):** Global surgical videos paired with entire abstract texts of the surgical procedure.

Disentangling Embedding Spaces

By maintaining separate projection heads for distinct semantic levels, HecVL prevents semantic "blurring" and facilitates zero-shot procedural tracking across unseen hospitals.

Handling Procedural Variations: Skipping Steps

The Challenge: Surgical procedures are dynamic. Different medical centers follow customized sequences, and steps are routinely skipped or substituted.

- **SurgPLAN++ (2024) Temporal Detection:**
  - Formulates recognition as temporal segment detection through phase proposals.
  - Uses data augmentation (mirroring, center-duplication, down-sampling) to extend active streaming video into "pseudo-complete" sequences.
  - Continuously refines preceding predictions at each online step, correcting temporal anomalies retrospectively.

Ensuring Temporal Consistency: StepAL & SFR

How do we ensure the tracker "stays on track" when a specific canonical step is missing?

- **Step-aware Feature Representation (SFR from StepAL):**
  - Models sequential dependencies by organizing clip embeddings according to predicted steps.
  - **Skip Mitigation:** If a step S_k is skipped, SFR pads the corresponding slot in the feature vector with the video's **global average feature representation**.
  - Preserves chronological dimensional consistency without losing the global workflow context.

Modular "Lego Recipe" Building

Are surgical steps modular?

- **The Hypothesis:** We can think of steps as modular, interchangeable "Lego bricks" across surgeries.
- **Modular Transfer:** Logging fine-grained steps within standardized macro-phases allows the alignment model to transfer concepts between surgeries (e.g., from cataract incision to laparoscopic bypass dissection).
- **Zero-Shot Transferability:** Enables generalist surgical AI that transfers across clinics without requiring manual retraining.

Proposed Methodology & Roadmap

1. **Step 1: Forward Parsing Pipeline**
   - Raw Video $\rightarrow$ Mask2Former (Segmentation) + DepthAnythingV3 (Centroid Depth) + Qwen3-VL (Actions) $\rightarrow$ ActDT JSON.
2. **Step 2: Spatio-Temporal Joint Aligner**
   - Aligns actual visual twins with hierarchical canonical drapes using HRAM cross-attention and Inter-Task Contrastive Learning (ICL).
3. **Step 3: Multi-Center Generalization Benchmarks**
   - Validate across three surgical specialties: Cataract (CAT-SG, Cataract-1k), Gastric Bypass (StrasBypass70/BernBypass70), and Prostatectomy (PSI-AVA).